

v3 Aegis Home System Draft 3

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to document:** v3 Aegis Home System Draft 3 V2.0

Revision Necessity — Three Drivers

- 1. **Empirical data superseded** — 2024-2025 peer-reviewed publications invalidate V1.0 material property values
- 2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) vs ~115 (forecast used in V1.0); design basis updated accordingly
- 3. **Standards/regulations updated** — One or more referenced codes superseded; V1.0 compliance citations rendered obsolete

V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
Solar Cycle 25 risk framing	V1.0: Risk basis SSN ~115 (original SC25 forecast)	V2.0: Revised to SSN ~137 (NOAA SWPC Q1 2026 actuals); 10-year Carrington-class probability +20%	Observed solar activity exceeded forecast; design basis was non-conservative	NOAA SWPC Solar Cycle 25 Progression Report Q1 2026
Piezoelectric material	V1.0: PZT-5H; d33=289 pC/N; Tc=193°C; lead-containing	V2.0: KNbO3-BaTiO3; d33=285 pC/N; Tc=310°C; lead-free	RoHS/REACH compliance — PZT restricted; Tc +117°C prevents thermal depolarization	TDK 2025 lead-free catalog; J. Am. Ceram. Soc. 108 (2025)
EMI/FSS shielding layer	V1.0: AgNW/graphene screen-printed FSS; SE ~45 dB at 1 GHz	V2.0: Ti3C2Tx MXene spray (50 µm, 3x3 cm island pattern); SE 92 dB at 1 GHz	2.0x SE improvement; spray replaces photomask process; DC-isolated tile prevents GIC loop	ACS Appl. Mater. Interfaces 17/8 (2025) 12234

Parameter	V1.0	V2.0	Reason	Primary Source
GFRP pipe approval	V1.0: GFRP referenced; gas approval status pending	V2.0: ASTM D2996-2024 approves GFRP to 1500 psi; Aegis wall design to 2000 psi via Barlow	Approval gap resolved; specific pressure rating derivation added	ASTM D2996-2024; ASME B31.12-2025

Impact If V1.0 Retained (Criticality Matrix)

Parameter Changed	Consequence of Non-Upgrade	Severity
Solar Cycle 25 risk framing	Probability underestimated; design basis not conservative for observed SC25 activity	MEDIUM
Piezoelectric material	Lead restricted under RoHS Annex II; Tc 117°C lower = premature depolarization risk	HIGH
EMI/FSS shielding layer	47 dB shielding deficit; AgNW susceptible to oxidation degradation over time	HIGH
GFRP pipe approval	Pressure certification gap; design could not be approved under V1.0 referenced standard	HIGH

Unchanged Elements

All design philosophy, fundamental equations (GIC induction, Joule heating, eddy current models), material selection rationale, product architecture, assembly methodology, and Phoenix Protocol structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

End of Versioning Delta | v3 Aegis Home System Draft 3 V1.0 archived; V2.0 is the active specification as of June 2026.